

ULAANBAATAR, Mongolia

WMO: 442920

Lat: 47.9183N

Lon: 106.8478E

Elev: 4279

StdP: 12.56

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.1	-29.2	-40.4	0.6	-32.6	-36.2	0.8	-29.2	18.7	-0.3	15.9	0.1	2.0	170	0.695

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	87.9	59.8	84.1	58.7	80.4	57.6	63.8	78.3	62.1	76.0	60.6	74.3	9.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.2	88.1	67.8	57.3	82.3	66.8	55.5	77.0	65.8	31.5	78.9	30.1	75.6	28.9	74.1	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	19.6	16.7	-36.9	93.7	7.0	4.5	-41.9	96.9	-46.0	99.6	-49.9	102.1	-55.0	105.4		
(5)			-36.9	67.3	7.0	2.4	-42.0	69.0	-46.0	70.5	-50.0	71.8	-55.1	73.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.7	-10.0	-1.0	17.8	36.1	49.0	61.0	66.1	61.3	49.0	32.1	10.1	-5.6		
	DBStd	28.02	8.43	10.77	12.08	9.46	9.11	7.76	6.43	6.63	8.32	8.94	12.35	9.22		
	HDD50	8446	1859	1429	1000	425	129	8	0	3	116	555	1198	1724		
	HDD65	12719	2324	1849	1465	867	499	163	61	152	482	1020	1648	2189		
	CDD50	1385	0	0	0	8	99	339	498	354	86	1	0	0		
	CDD65	183	0	0	0	0	5	45	94	38	1	0	0	0		
	CDH74	2735	0	0	0	4	179	737	1142	594	79	0	0	0		
	CDH80	879	0	0	0	0	39	234	416	180	10	0	0	0		
	WSAvg	5.8	3.0	3.9	5.8	8.0	8.6	7.7	6.8	6.3	6.4	5.5	4.0	3.0		
	PrecAvg	10.0	0.1	0.1	0.2	0.3	0.9	1.6	2.8	2.4	1.0	0.4	0.2	0.2		
(15) Precipitation	PrecMax	14.8	0.3	0.2	0.8	0.7	2.5	3.3	6.3	5.6	2.8	1.3	0.6	0.3		
	PrecMin	7.0	0.0	0.0	0.0	0.0	0.1	0.2	0.3	0.5	0.0	0.1	0.0	0.0		
	PrecStd	1.9	0.1	0.1	0.2	0.2	0.7	0.7	1.2	1.1	0.7	0.3	0.2	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.6	33.6	55.4	71.7	84.7	91.7	94.8	91.0	80.9	65.6	45.0	21.8		
		MCWB	16.5	25.8	38.7	46.7	53.9	59.3	63.9	61.5	54.2	45.6	33.9	18.4		
	2%	DB	13.4	27.0	48.0	65.2	78.9	86.1	89.2	84.5	75.5	59.1	39.0	16.9		
		MCWB	10.9	22.1	35.0	44.8	52.0	57.6	61.7	59.0	53.7	42.8	30.7	14.4		
	5%	DB	8.8	22.3	42.7	60.5	74.0	82.0	84.4	80.6	71.4	54.0	33.6	13.0		
		MCWB	7.1	18.5	32.0	42.0	50.1	56.3	60.7	58.4	51.8	40.2	27.5	11.1		
	10%	DB	4.7	17.5	37.1	55.6	69.5	77.8	80.4	76.6	67.0	49.5	28.7	9.4		
		MCWB	3.2	14.5	29.0	39.8	48.4	55.7	59.9	58.1	50.3	37.9	24.3	7.9		
	0.4%	WB	16.3	26.6	39.5	48.8	56.4	62.9	67.1	65.2	59.0	46.4	34.7	18.7		
		MCDB	19.1	32.6	53.8	68.9	78.3	77.7	82.4	79.7	70.8	62.8	44.4	21.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	10.8	22.3	35.3	45.4	53.5	60.8	64.7	62.8	56.0	43.4	30.9	14.4		
		MCDB	13.1	27.1	46.5	63.8	75.3	76.2	79.8	76.1	69.9	57.9	38.1	16.6		
	5%	WB	7.1	18.4	32.2	42.7	51.1	59.2	63.2	61.0	53.7	40.7	27.6	11.2		
		MCDB	8.7	22.0	41.7	58.7	71.1	74.5	77.6	74.2	68.1	53.2	33.2	13.0		
	10%	WB	3.2	14.5	29.2	40.1	48.9	57.7	61.7	59.4	51.4	38.1	24.5	7.9		
		MCDB	4.6	17.5	36.8	54.5	67.4	72.9	75.7	71.9	64.8	48.4	28.6	9.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	24.4	28.5	27.5	27.8	28.0	24.7	23.2	24.1	27.8	27.3	24.3	22.9		
		MCDBR	27.1	32.3	32.6	37.0	37.7	32.5	30.6	33.0	35.8	35.9	28.2	24.0		
		MCWBR	24.9	27.7	21.1	20.7	18.2	12.8	11.3	13.3	17.5	21.5	20.8	21.9		
		MCDBR	27.0	31.8	30.5	34.2	33.0	25.7	24.1	25.6	30.3	33.7	27.3	23.8		
	5% WB	MCWBR	24.9	27.4	20.0	19.5	16.8	11.5	10.4	11.6	15.4	20.4	20.3	21.8		
(40) Clear-Sky Solar Irradiance	taub		0.241	0.264	0.304	0.351	0.382	0.392	0.377	0.357	0.313	0.284	0.258	0.243		
	taud		2.336	2.291	2.268	2.237	2.190	2.265	2.358	2.389	2.456	2.472	2.396	2.329		
	Ebn at Noon		272	287	290	285	279	276	279	280	285	278	262	255		
	Edh at Noon		22	29	35	41	44	42	37	35	29	24	21	19		
(44) All-Sky Solar Radiation	RadAvg		613	979	1428	1794	2063	2056	1937	1720	1444	998	632	482		
	RadStd		33	58	60	76	147	115	109	115	79	50	34	30		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	-4.14	-5.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page